

vimpute(): An Extension of the Imputation Methods in the VIM Package

Eileen Vattheuer

Eileen.Vattheuer@statistik.gv.at

Nina Niederhametner

Nina.Niederhametner@statistik.gv.at

Alexander Kowarik

Alexander.Kowarik@statistik.gv.at

Johannes Gussenbauer

Johannes.Gussenbauer@statistik.gv.at

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www.statistik.at

R-Package VIM

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- VIM: **V**isualization and **I**mputation of **M**issing Value

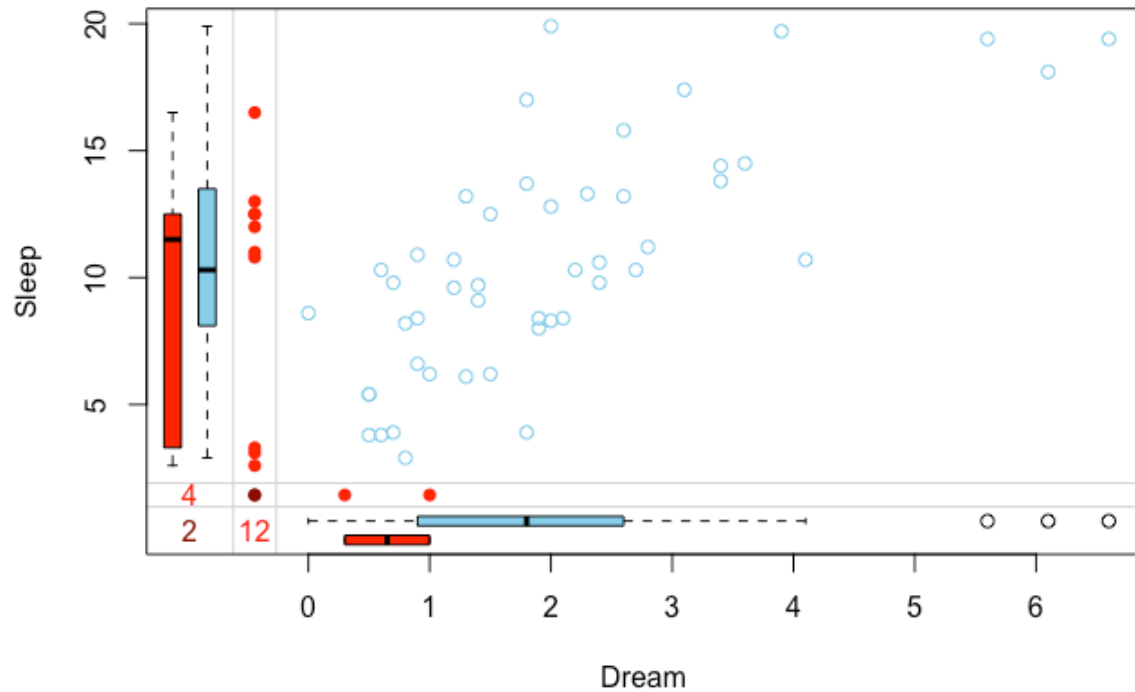
- Developed for the use of tabular data
 - Records ~ rows
 - Variables ~ columns

- Contains various imputation methods

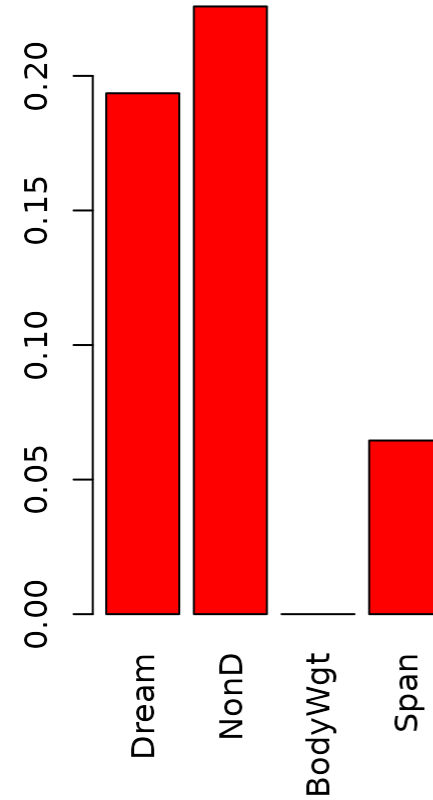
- Available on CRAN and actively developed
 - <https://cran.r-project.org/web/packages/VIM/index.html>
 - <https://github.com/statistikat/VIM>

$$\mathbf{X} = \begin{pmatrix} x_{11} & \dots & \dots & x_{1p} \\ \vdots & \text{NA} & & \vdots \\ & & & \text{NA} \\ \vdots & & \text{NA} & \vdots \\ x_{n1} & \dots & \dots & x_{np} \end{pmatrix}$$

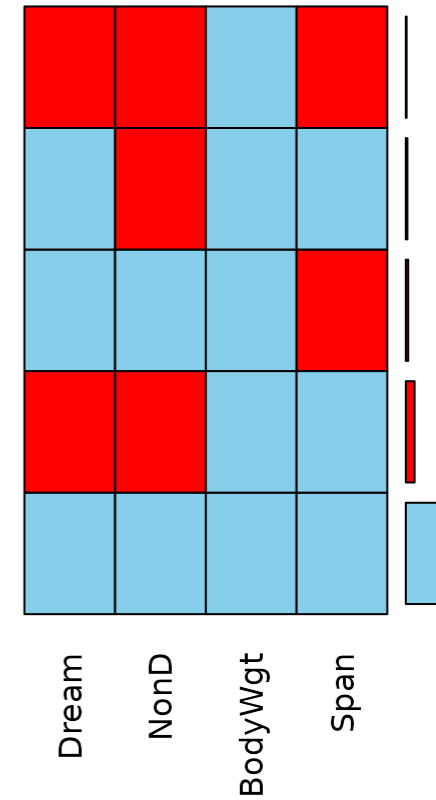
Visualization of Missing Values



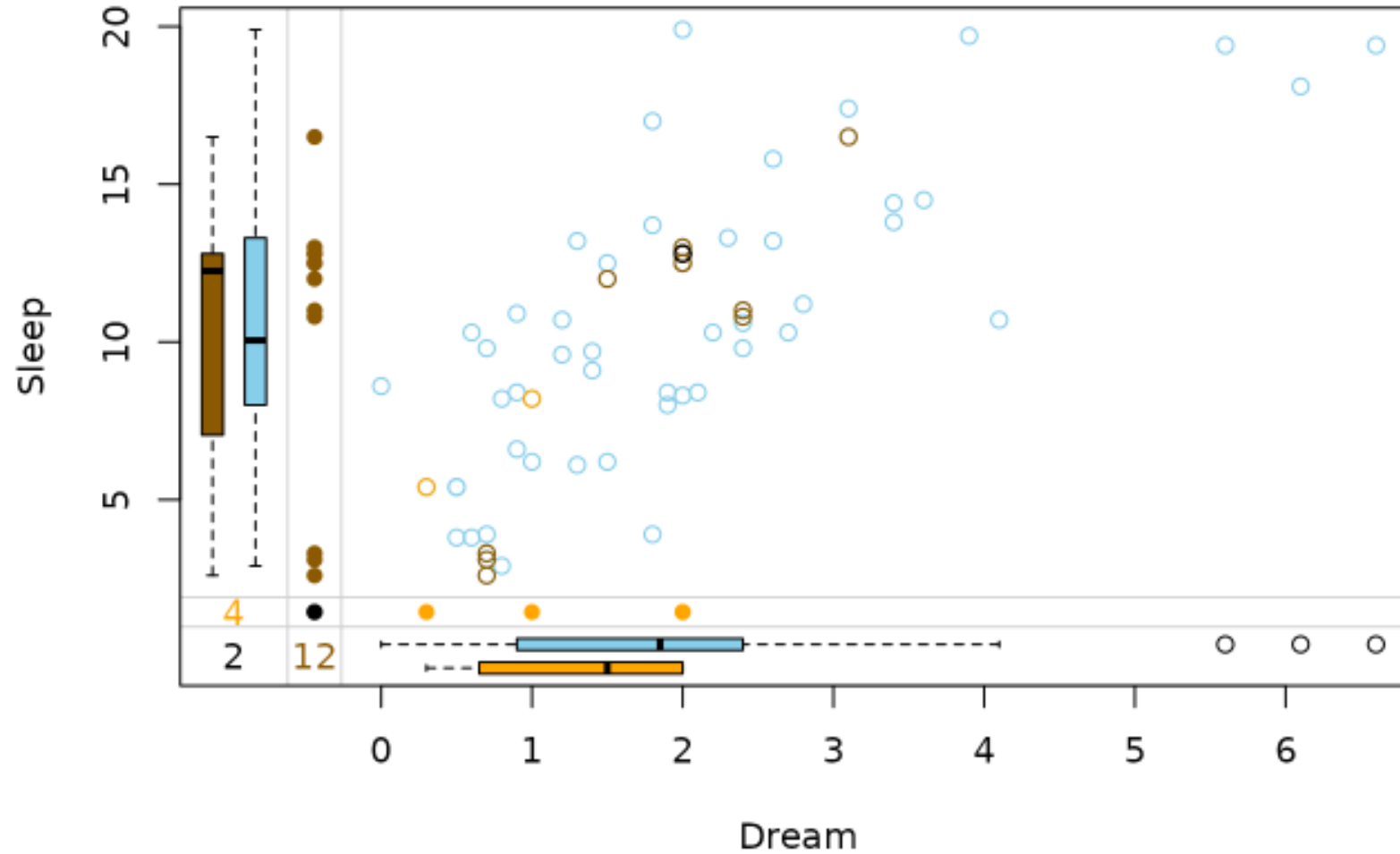
Proportion of missings



Combinations



Visualization of Imputed Values



Imputation Methods

Donor based methods

```
kNN(data, variable = ..., k = 5, dist_var = ..., ...)
```

```
hotdeck(data, variable = ..., ord_var = ..., domain_var = ..., ...)
```

```
matchImpute(data, variable = ..., match_var = ..., ...)
```

Model based methods

```
regressionImp(formula = ..., data = ...family = ...,
```

```
# Iterative robust model-based imputation
```

```
irmi(x = data, maxit = 100, noise = ..., robust = ..., ...)
```

Imputation Methods

ML based methods

```
rangerImpute(formula = ..., data = ..., ...)
```

```
xgboostImpute(formula = ..., data = ..., ...)
```

```
transformerImpute(data = ..., target = ..., cat_vars = NULL, ...)
```

VIM – Strengths and Limitations

Strengths:

- Wide range of imputation methods: kNN, hotdeck, regressionImp, irmi
- Easy to use
- Transparent diagnostics: visualization of missing data patterns
- Iterative/sequential imputation available (irmi)

Limitations:

- Each method must be called individually → no unified framework
- No variable-specific model selection
- Sequential iterations limited to irmi → not available for other methods
- Predictive Mean Matching (PMM) only partially available



`vimpute()`

Key Features and Advantages of vimpure()

- Unified framework
- Single function to handle imputation of multiple variables
- Variable-specific models
- PMM
- Automatic iteration & convergence check
- Supports modern ML models via mlr3
- Hyperparameter tuning possible
→ finds best model settings automatically

Age	Education	Yearly Income
30	Post-Secondary	NA
NA	Secondary	28000
52	Tertiary	NA
18	NA	0

↓
**robust
regression**

↓
**random forest
+
tuning**

↓
**xgboost
+
PMM**

Workflow

Check data

For n iterations:

For each variable with missings in the dataset:

- **Create training task**: which data, target variable, task type
- **Create learner**: choose a machine learning model
- **Hyperparameter tuning** (optional)
- Handling of missing values
- **Train model**: `learner$train(train_task)`
- **Create prediction task**
- **Predict missing values**: `learner$predict(pred_task)`
- Apply **PMM** (optional)
- **Replace** missing values in dataset

Check stopping criteria

Return final dataset

semi-continuous
vs. continuous

Parameters

```
vimpute(  
  data = as.data.table(sleep),  
  considered_variables = c("BodyWgt", "BrainWgt", "Dream"),  
  method = list(BodyWgt = "ranger",  
                 BrainWgt = "xgboost",  
                 Dream = "robust"),  
  pmm = list(BodyWgt = TRUE,  
              BrainWgt = FALSE,  
              Dream = FALSE),  
  formula = FALSE,  
  nseq = 10,  
  tune = FALSE )
```

mlr3 Integration

- Modular, principled framework: Task, Learner, Tuning
- Unified interface: one consistent interface for all models for different ML models
- Large & extensible ecosystem
- Integrated hyperparameter tuning
- Parallelization support

→ **Structured, flexible, modern ML framework with tuning and parallelization built-in**

Methods

ranger – random forest 

xgboost – extreme gradient boosting 

regularized – regularized lm 

robust – robust linear regression 

- Not directly from mlr3
- Wraps `robustbase::lmrob()` in an mlr3-compatible `LearnerRegr` object
- Inspired by `irmi`

Simulation Study

Simulation Study

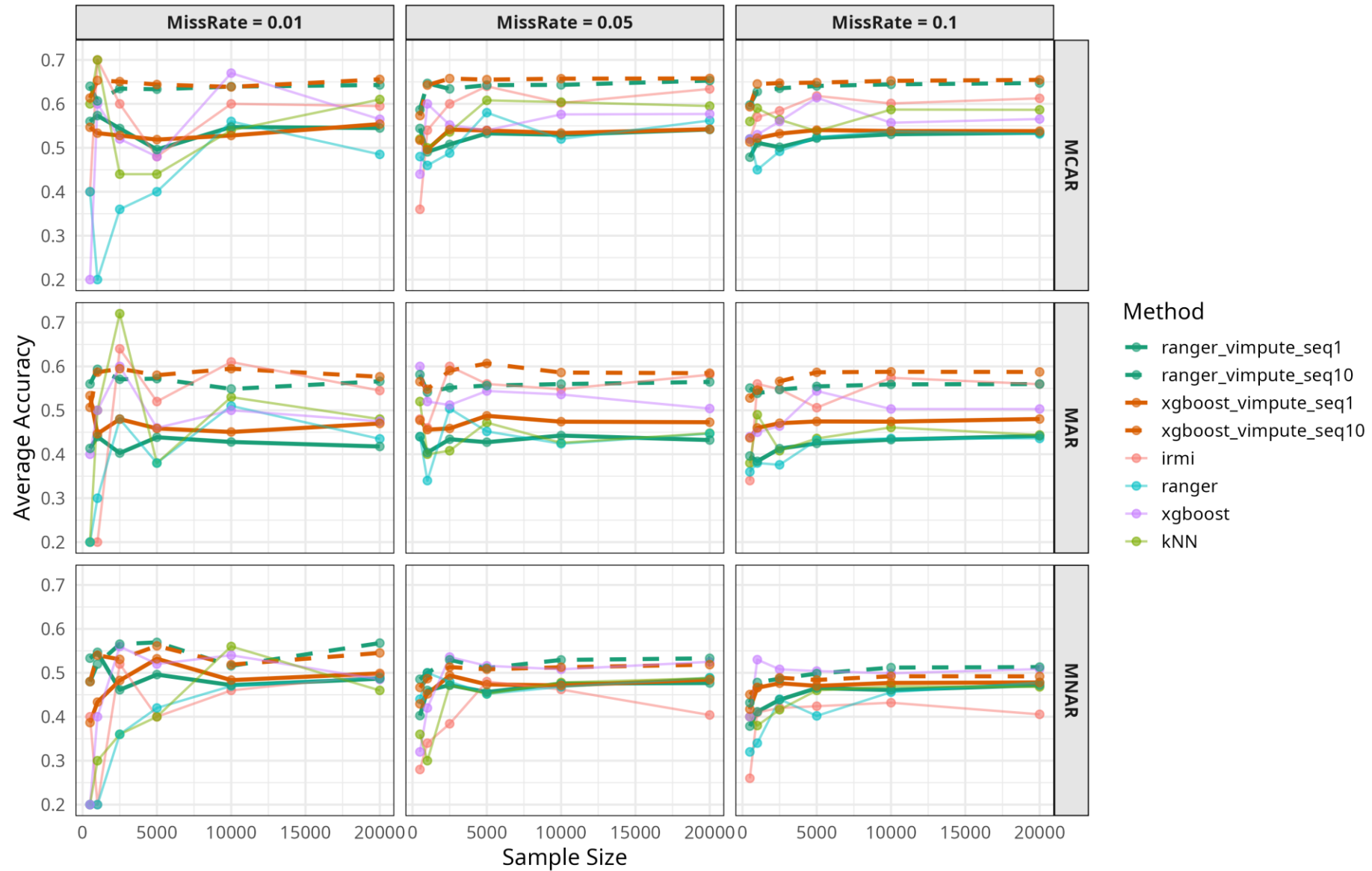
- Aim: Test methods from new framework
- Data: Richframe ~ housing register containing all registered persons in Austria living in private households
 - Variety of variable: Geographic variables, variables on household structure, sociodemographic variables, ...

HID	NUTS2	age	sex	Citizenship	Education	Yearly Income
1	AT13	30	m	AT	Post-Secondary	25000
2	AT32	56	m	EU	Secondary	28000
2	AT32	52	f	AT	Tertiary	32000
2	AT32	18	f	AT	Secondary	0

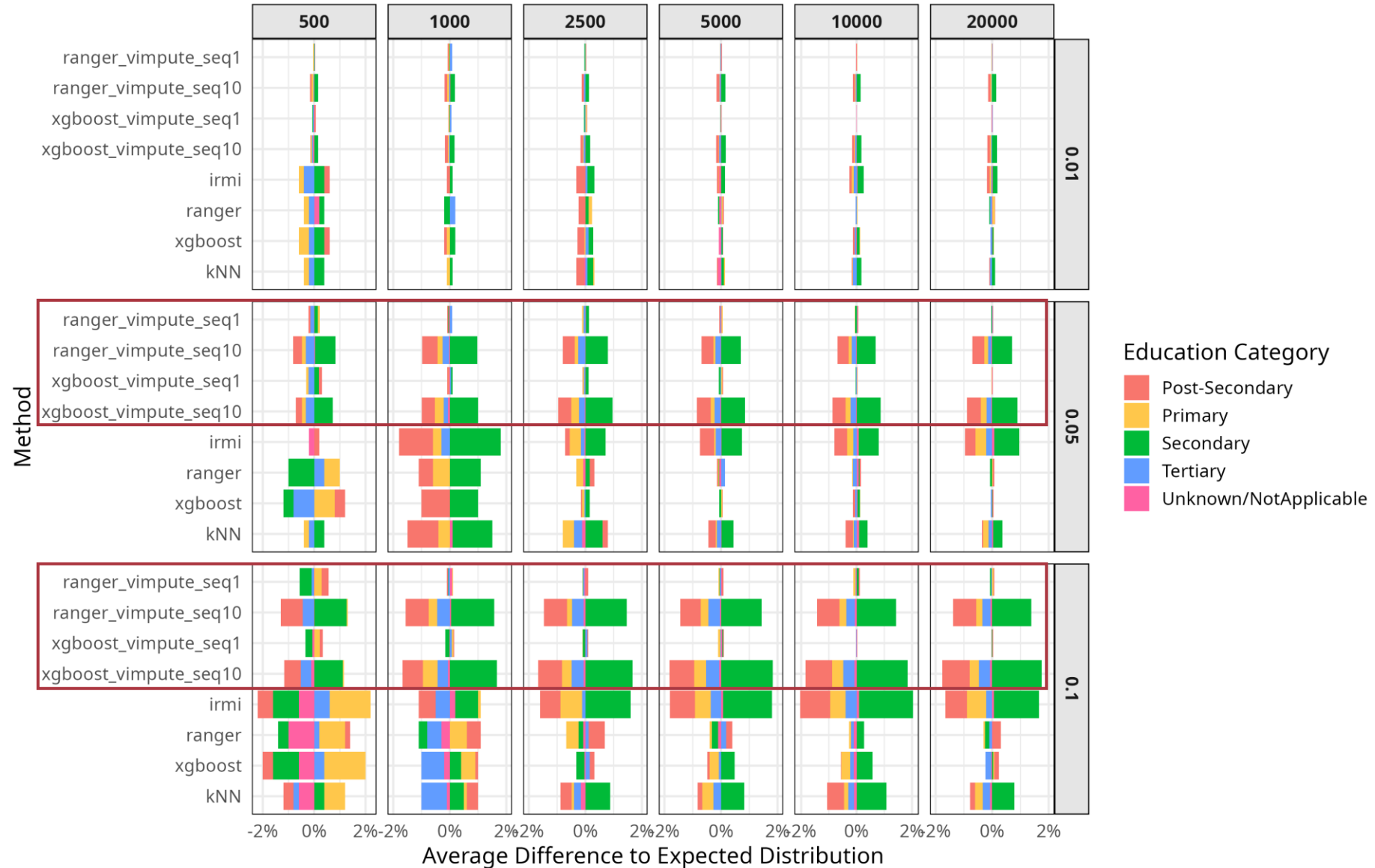
Simulation Study

- Take sample from Richframe → add missing values for a specific variable → apply imputation method → compare results
- Apply different missing mechanisms
 - MCAR: randomly draw position of missing values
 - MAR: Missingness depends on other observed variables
 - MNAR: Missingness depends directly on the value of the variable itself.
- Simulate MAR or MNAR we derived occurrence of missing value from typical non response patterns
 - Higher response rates: rural areas, higher education, higher yearly income
 - Lower response rates: urban areas, lower education, migration background, low or very high income

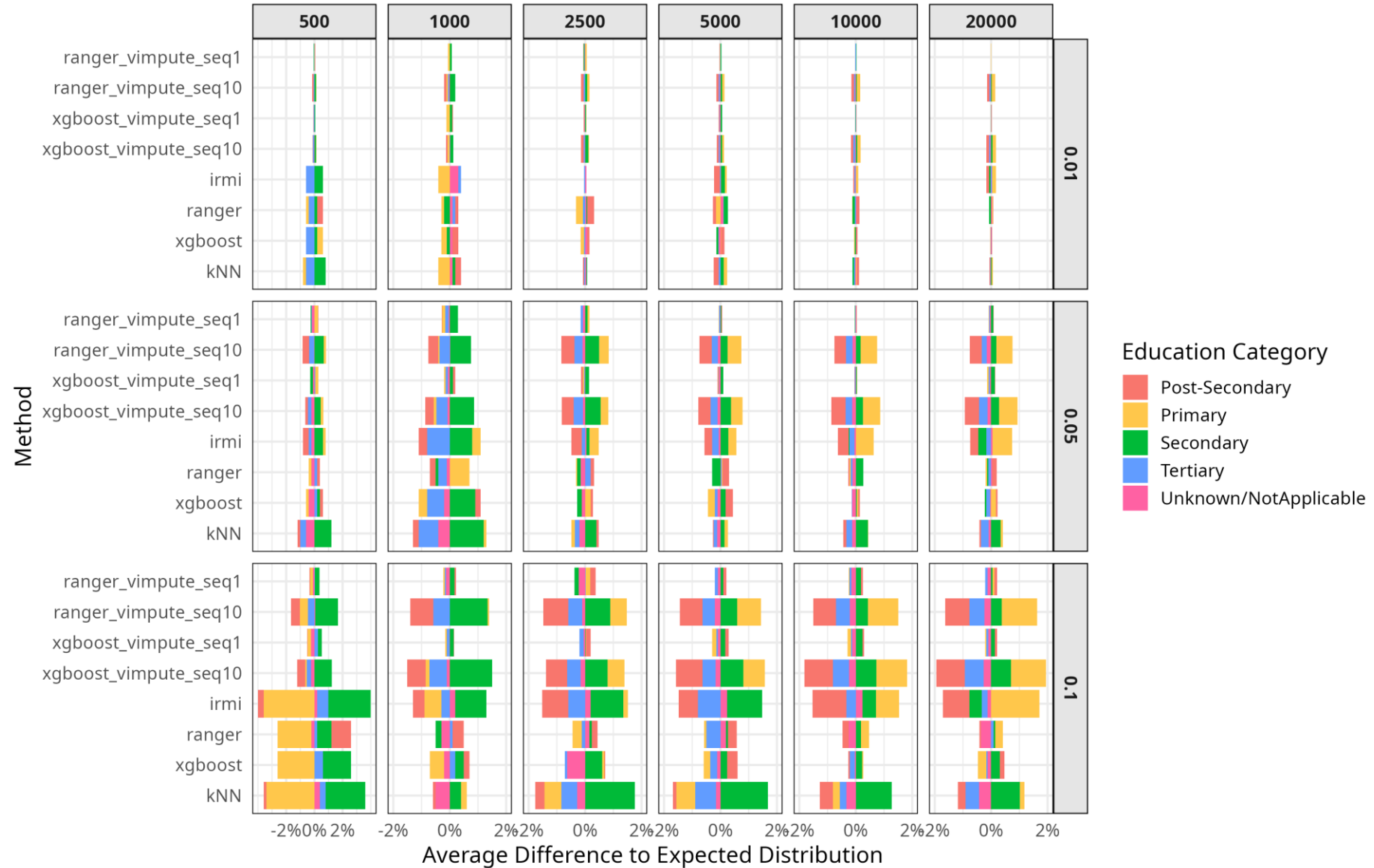
Average Accuracy - Education



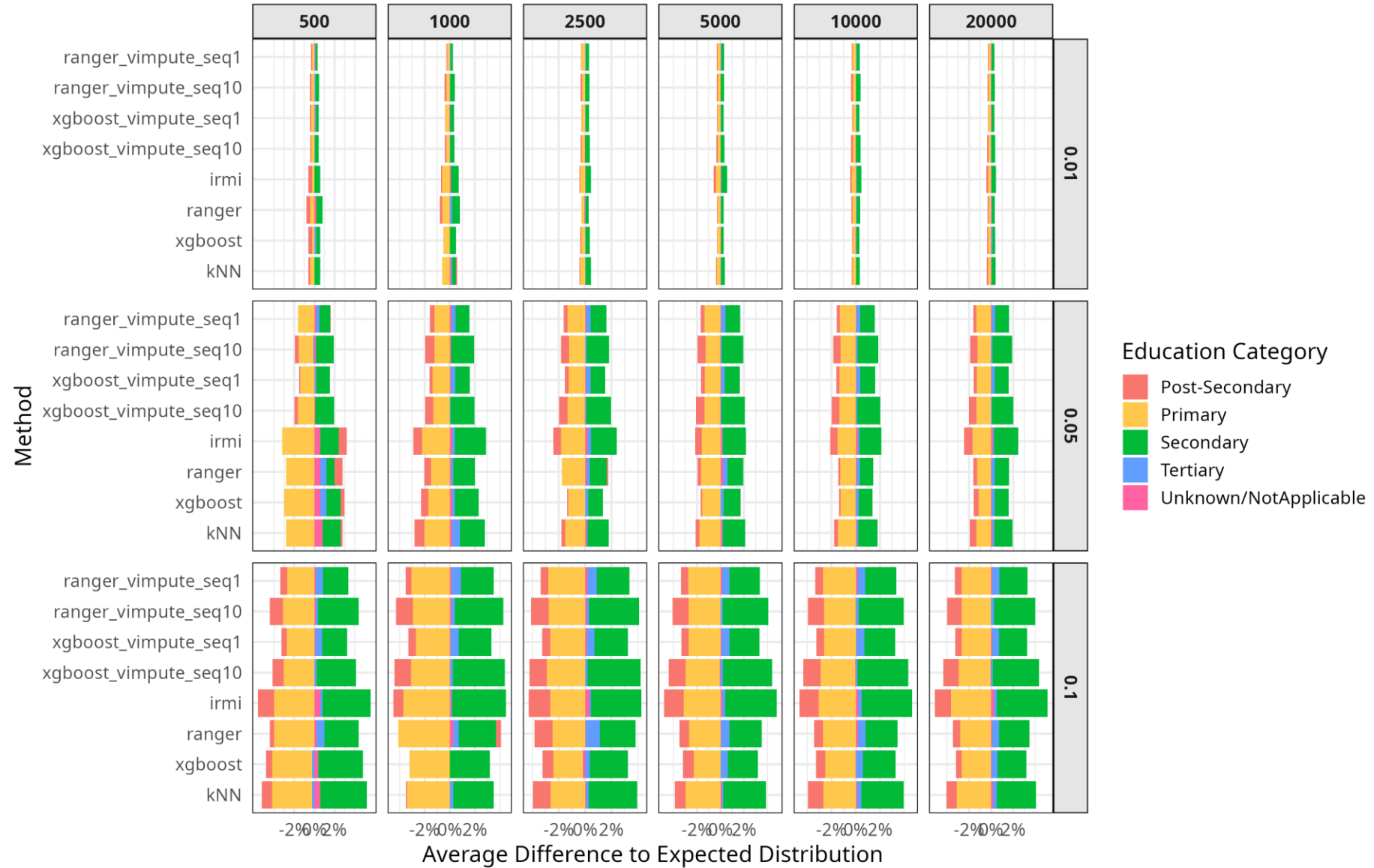
MCAR - Education



MAR - Education



MNAR - Education



Conclusion

Good approach to bring everything into one unified framework

Expandable framework

Trade off between prediction performance and to preserve the dataset's distributions

Please address queries to

Eileen Vatheuer

Eileen.Vatheuer@statistik.gv.at

Any Questions?

STATISTICS AUSTRIA

Guglgasse 13, 1110 Vienna

 statistik_at

 statistik_at

 statistics-austria

 @statistik_austria

 statistik.at